OPERATIONS & ADMINISTRATION MANUAL

Vehicle Zone Intelligence — User Manual

Airport Arrival Ramp Monitoring Dashboard

Contents

About this manual

This manual documents every screen and every parameter of the **Vehicle Zone Intelligence** dashboard — the web application that monitors vehicle occupancy, dwell time, and alerts across the airport arrival ramp.

It is organised into two parts:

Part	Audience	Covers
Part A — User (Operations)	Operators / ramp controllers	Overview, Live View, Alerts, Analytics, the notification bell, and the global shell. Day-to-day monitoring.
Part B — Admin (Configuration)	Administrators on the server console	Cameras, Zones, Edge Status, and the Live Data Feed. System setup and health.

> Every red box / ellipse in the screenshots marks a parameter. The number in the red tag matches the numbered explanation underneath the image.

How to open the dashboard

Open a browser and go to:

```
https://localhost/dashboard
```

The application opens on the **Overview** page. You can jump straight to a page by adding its name after a `#`, for example `https://localhost/dashboard#alerts`.

Two access levels (Admin vs Viewer)

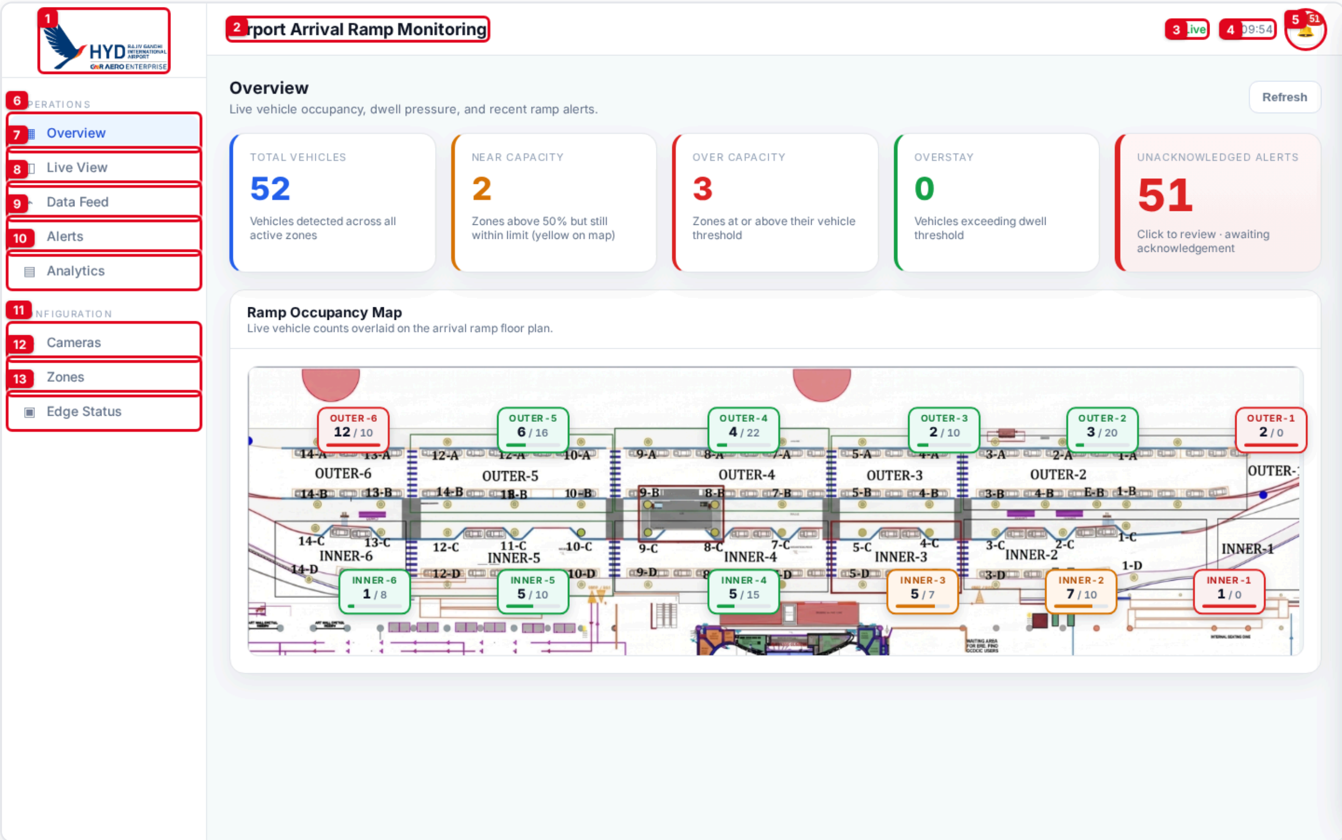
The dashboard decides what you can see from **where you connect**:

- **Admin** — the browser running **on the server console (localhost)**. Sees everything, including the **Configuration** pages (Cameras, Zones, Edge Status) and the **Data Feed**.
- **Viewer** — any other client on the network. Sees the **Operations** pages only. The Configuration section and the Data Feed are hidden from the sidebar and blocked even if reached by direct link.

This is why the **Data Feed** is documented in Part B (Admin): it is an admin-only diagnostic surface, even though the page sits under the "Operations" heading in the sidebar.

The global shell (header & navigation)

Every page shares the same frame: a left **navigation rail** and a top **header bar**.



Header bar (top):

1. **Brand logo** — airport / operator branding (Rajiv Gandhi International Airport · GMR Aero Enterprise). Clicking it does nothing; it is an identifier.
2. **Page / system title** — "Airport Arrival Ramp Monitoring". Fixed banner for the whole deployment.
3. **Connection state** — live WebSocket status. **Live** (green dot) = real-time data is streaming; **Connecting** = trying to (re)connect; red = connection error. If this is not "Live", numbers on the page may be stale.
4. **Clock** — the operator workstation's local time, updated every second.
5. **Notification bell** — opens the *Recent Alerts* drawer (see Notification bell & drawer). A red **badge** on the bell shows the number of **unacknowledged** alerts (shows `99+` above 99). The bell gently pulses while unacknowledged alerts exist.

Navigation rail (left) — Operations:

1. **Overview** — ramp-wide KPIs and the live occupancy map.
2. **Live View** — annotated camera streams.
3. **Data Feed** — raw message stream *(admin-only; hidden for viewers)*.
4. **Alerts** — the full operational alert log.

5. **Analytics** — historical alert trends.

Navigation rail (left) — Configuration *(admin-only; the whole section is hidden for viewers):*

1. **Cameras** — RTSP camera sources.
2. **Zones** — ramp geometry, capacity and dwell rules.
3. **Edge Status** — edge device telemetry.

> On phones the rail collapses into a ☰ **hamburger** menu, and the Configuration entries are hidden regardless of access level.

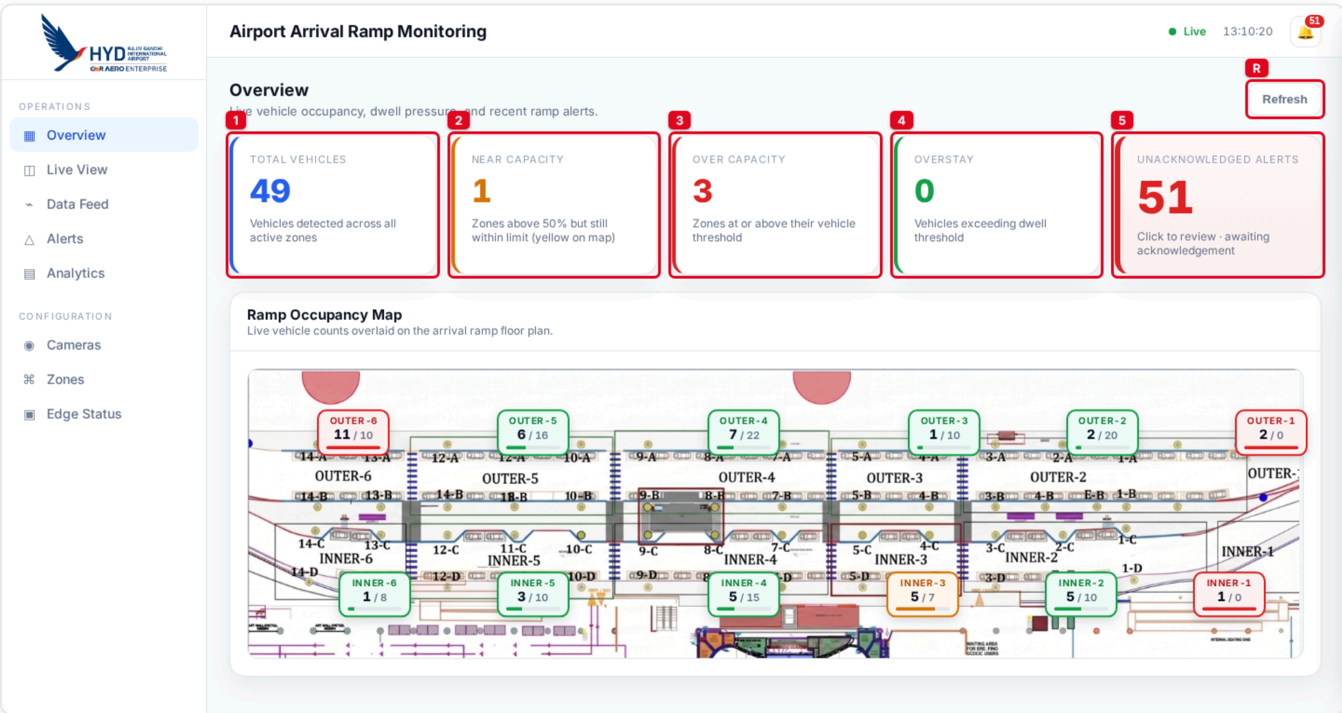
SECTION

Part A — User (Operations)

1. Overview

`#overview` — the default landing page. It answers "how busy is the ramp right now, and is anything wrong?" at a glance.

1.1 Key Performance Indicators (the five cards)



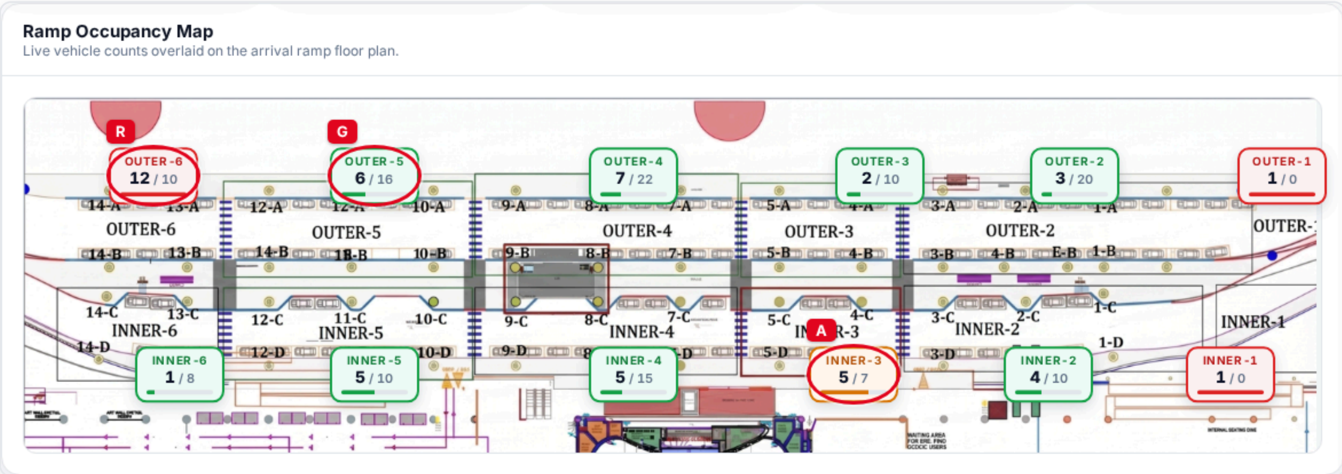
R. Refresh — forces an immediate re-fetch of all overview data. (The page also updates itself live over the WebSocket; this button is for an on-demand pull.)

1. **Total Vehicles** *(blue)* — the total number of vehicles currently detected across **all active zones**. Multi-camera zone *groups* are summed once, so a single physical zone watched by four cameras is not counted four times.
2. **Near Capacity** *(amber when > 0, else green)* — how many zones are **above 50 % occupancy but still within their limit** (the "Busy"/yellow tiles on the map below). A zone counted here is *not* also counted as Over Capacity.
3. **Over Capacity** *(red when > 0, else green)* — how many zones have **more vehicles than their Max Vehicles threshold** (strictly over the limit). These are the conditions that raise *overcrowding* alerts.
4. **Overstay** *(amber when > 0, else green)* — the total number of vehicles that have **exceeded their zone's dwell-time threshold** (parked / waiting too long). These raise *overstay* alerts.
5. **Unacknowledged Alerts** *(red when > 0, else green)* — the count of alerts still **awaiting acknowledgement**. This card is **clickable** — click it to jump straight to the **Alerts** page. The same number drives the header bell badge.

> **Colour logic for occupancy throughout the app:** ≤ 50 % = **green** (Calm),
>
> > 50 % up to the limit = **amber** (Busy), and strictly over the limit = **red** (Over capacity).

1.2 Ramp Occupancy Map

A live floor plan of the arrival ramp. The top row is the **OUTER** ramp (OUTER-6 → OUTER-1, left to right); the bottom row is the **INNER** ramp (INNER-6 → INNER-1). Each slot shows a tile ("chip") with the live count.



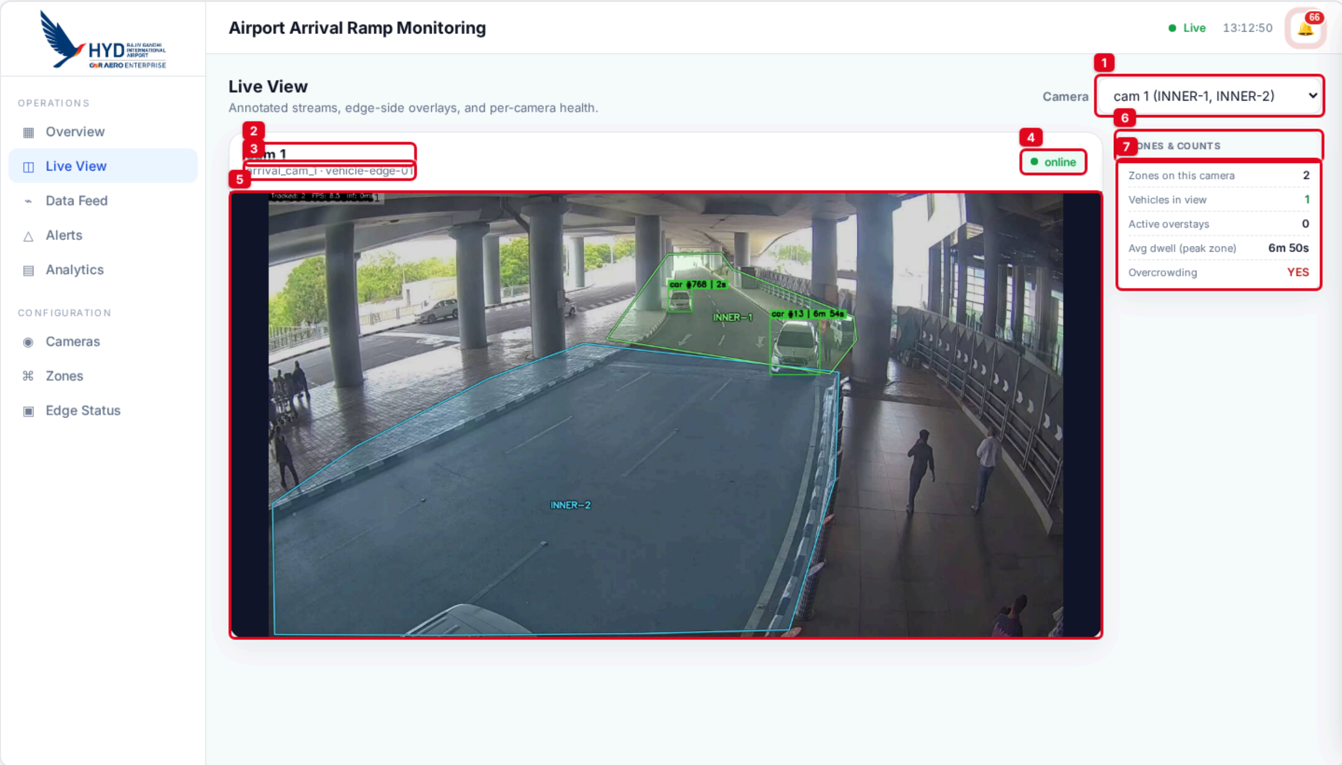
Each chip shows `<current count> / <Max Vehicles>` and a small fill bar, and is colour-coded by occupancy:

- **R — Red chip (Over capacity):** count is over the zone's limit (e.g. `OUTER-6 12 / 10`). Matches the *Over Capacity* KPI and overcrowding alerts.
- **A — Amber chip (Busy):** above 50 % but still within the limit. Matches the *Near Capacity* KPI.
- **G — Green chip (Calm):** at or below 50 % occupancy.
- **M — Muted / grey dashed chip (–):** a ramp slot that exists on the floor plan but has **no monitored zone** mapped to it. Shown so the plan stays visually complete. (Hover any chip for an exact `count/max (percent)` tooltip.)

> A zone only appears on this map if its **Map slot** name is exactly one of `OUTER-1` … `OUTER-6` or `INNER-1` … `INNER-6` (case-sensitive). See [Zones] for how that is configured.

2. Live View

`#live` — annotated camera streams with edge-side overlays and per-camera health.



1. **Camera selector** — choose which camera to watch. Each entry shows the camera name and the zones it covers, e.g. `cam 1 (INNER-1, INNER-2)`.
2. **Stream title** — the friendly name of the selected camera (e.g. `cam 1`).
3. **Stream meta** — the camera's technical ID and the edge runtime it is assigned to, e.g. `arrival_cam_1 · vehicle-edge-01`.
4. **Camera status chip** — `online` (green) / `offline` / `error` / `warning`, reflecting whether the edge is currently receiving frames.
5. **Annotated video frame** — the live picture with the edge's analytics drawn on top: **zone polygons** (the shaded/green outline of each monitored area) and **detection boxes** around each detected vehicle.

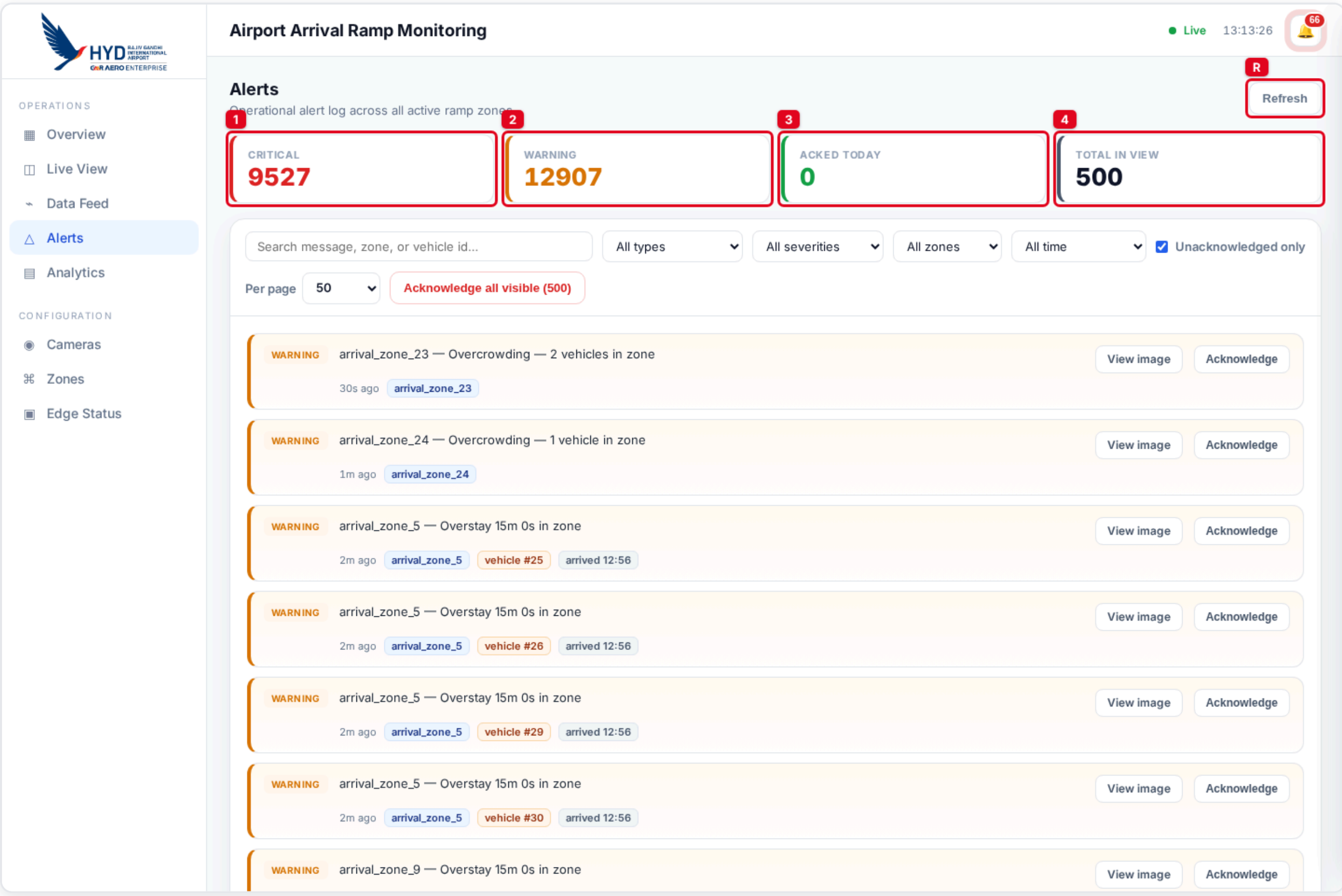
Side panel — "Zones & counts" (right). A live roll-up for the selected camera:

1. **Panel heading** — "Zones & counts".
2. **Per-camera metrics**, each on its own row:
3. **Zones on this camera** — how many monitored zones this camera covers.
4. **Vehicles in view** — total vehicles currently detected across those zones (green when > 0).
5. **Active overstays** — vehicles over their dwell threshold right now (amber when > 0).
6. **Avg dwell (peak zone)** — the average dwell time of the busiest zone on this camera, e.g. `6m 50s`.
7. **Overcrowding** — **YES** (red) if any zone on this camera is over its limit, otherwise **no** (grey).

3. Alerts

`#alerts` — the complete operational alert log across all ramp zones. This is the main page operators act from. The screen has three parts: summary cards, a filter toolbar, and the alert list.

3.1 Summary cards

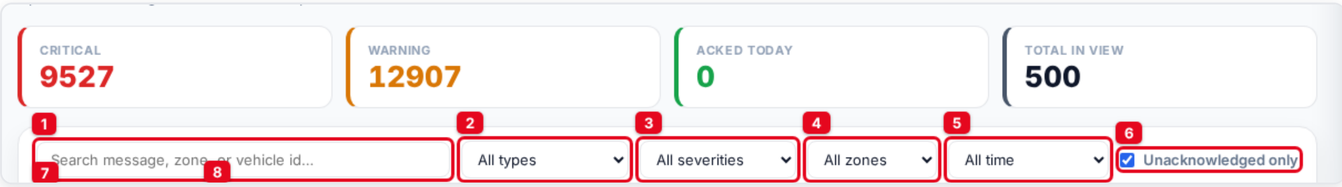


These reflect the **full** set matching the current type/zone filter (not just the visible page), so the severity counts are always true.

1. **Critical** *(red)* — number of critical-severity alerts.
2. **Warning** *(amber)* — number of warning-severity alerts.
3. **Acked today** *(green)* — number of alerts that have been acknowledged.
4. **Total in view** — total number of alerts matching the current filters.

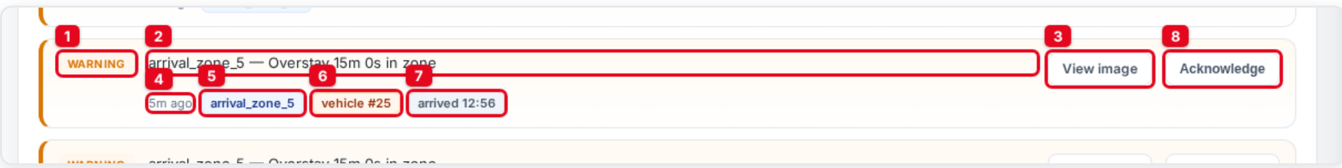
R. Refresh (top-right of the page) re-pulls the alert log from the server.

3.2 Filter toolbar



1. **Search** — free-text search over message text, zone, camera, and vehicle ID.
2. **Type filter** — *All types*, **Overcrowding**, **Overstay**, or **Camera offline**.
3. **Severity filter** — *All severities*, **Critical only**, or **Warning only**. (Picking a severity also sorts critical-first; otherwise the list is newest-first.)
4. **Zone filter** — limit to a single zone/zone-group by its friendly label (e.g. `OUTER-5`). The list is built from your configured zone names.
5. **Time filter** — *All time*, or the last **3 / 6 / 12 / 24 hours**.
6. **Unacknowledged only** — checkbox (on by default) to hide alerts that have already been acknowledged.
7. **Per page** — page size for the list: **10 / 50 / 100** (default 50).
8. **Acknowledge all visible** — acknowledges every un-acknowledged alert in the current filtered view in one click. The button shows the count (e.g. *Acknowledge all visible (500)*) and is disabled when there is nothing to acknowledge.

3.3 Anatomy of an alert row



1. **Severity badge** — **CRITICAL** (red left border) or **WARNING** (amber). Acknowledged rows are dimmed and lose their colour accent.
2. **Message headline** — always led by the live **zone label**, then a plain-language description:
3. *Overstay* → `… — Overstay 15m 0s in zone` (and `, N over limit` when past the threshold).
4. *Overcrowding* → `… — Overcrowding — N vehicles in zone`.
5. *Camera offline/recovered* → the maintenance message verbatim.
6. **View image** — opens the **evidence photo** for this alert in a new tab (a vehicle snapshot for overstay, or a frame snapshot for overcrowding). Shown only when an image exists.
7. **Time** — how long ago the alert fired (`5m ago`, `2h ago`, …). Hover for the exact timestamp.
8. **Zone chip** *(blue)* — the zone (or zone group) the alert belongs to. Hover to see the raw zone ID.
9. **Vehicle chip** *(orange)* — the tracked vehicle's ID, e.g. `vehicle #25` (present when the alert is tied to a specific vehicle).

10. **Arrived chip** — *overstay only* — when the vehicle entered the zone, e.g. `arrived 12:56`.
11. **Acknowledge** — marks this single alert as handled. The row then dims and shows `✓ Acknowledged`, the bell badge and *Unacknowledged Alerts* KPI drop by one. (Acknowledging never deletes an alert; it just clears it from the active queue.)

> **Toasts.** When a new alert arrives, a pop-up toast slides in at the top-right (up to three at once, auto-dismiss after ~5 s, red for critical / amber for warning). Click a toast to open the notification drawer.

3.4 When does an alert fire? (timing & frequency)

Alerts are **not** raised the instant a number crosses a line — both alert families use deliberate timing so a brief blip does not spam the log, and a sustained problem keeps escalating.

Overcrowding (per zone / zone-group):

- A zone must stay **over its Max Vehicles limit continuously for 2 minutes** before the **first** overcrowding alert fires. A spike that clears within 2 minutes raises nothing.
- While the zone stays over the limit, a **repeat** alert fires **every 15 minutes**.
- The moment the count drops back to / below the limit, the timer resets — the next breach must again be sustained for 2 minutes before it alerts.

Over-limit duration	Alert
0 - 2 min	(none - debounce)
2 min	1st alert
17 min	2nd alert
32 min	3rd alert
every +15 min while still over	4th, 5th ...

Overstay (per vehicle - an escalating "milestone ladder"):

Overstay milestones are multiples of the zone's **Max Dwell** threshold **T** (default `T = 15 min`). Each milestone fires **exactly once per visit**, carrying the vehicle's current (growing) dwell time, and escalates in severity. Because they scale with T, the cadence stays correct if you reconfigure a zone's dwell limit.

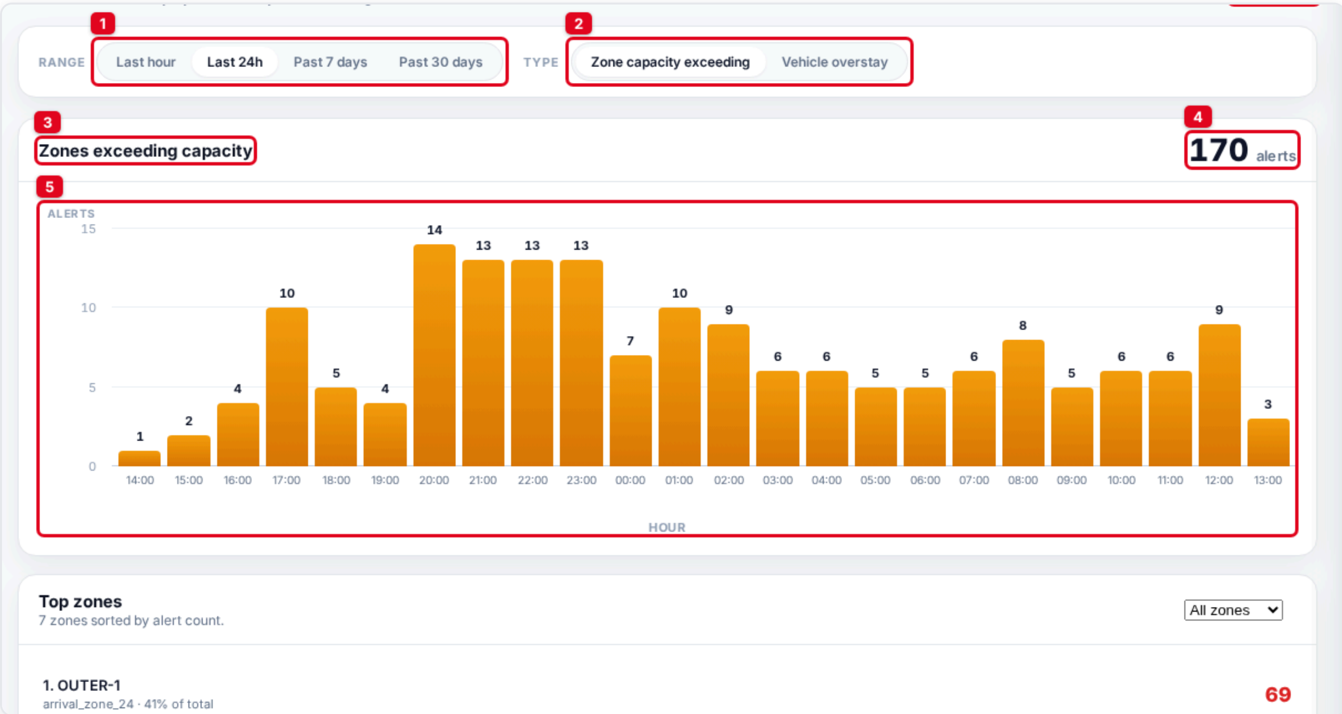
#	Fires at	Default (T = 15 min)	Severity
1st	1 x T	15 min	warning
2nd	2 x T	30 min	warning
3rd	4 x T	60 min	critical
4th	8 x T	120 min	critical
5th	12 x T	180 min	critical
...	every +4 x T	every +60 min (hourly)	critical

> Notes: with the defaults there is a deliberate gap between the 30-minute warning and the 60-minute first critical (nothing fires at 45 min). After the first critical, criticals repeat hourly for as long as the vehicle stays. The ladder resets when the vehicle leaves the zone, and the milestone already reached is remembered (even across an edge restart) so a step never re-fires.

4. Analytics

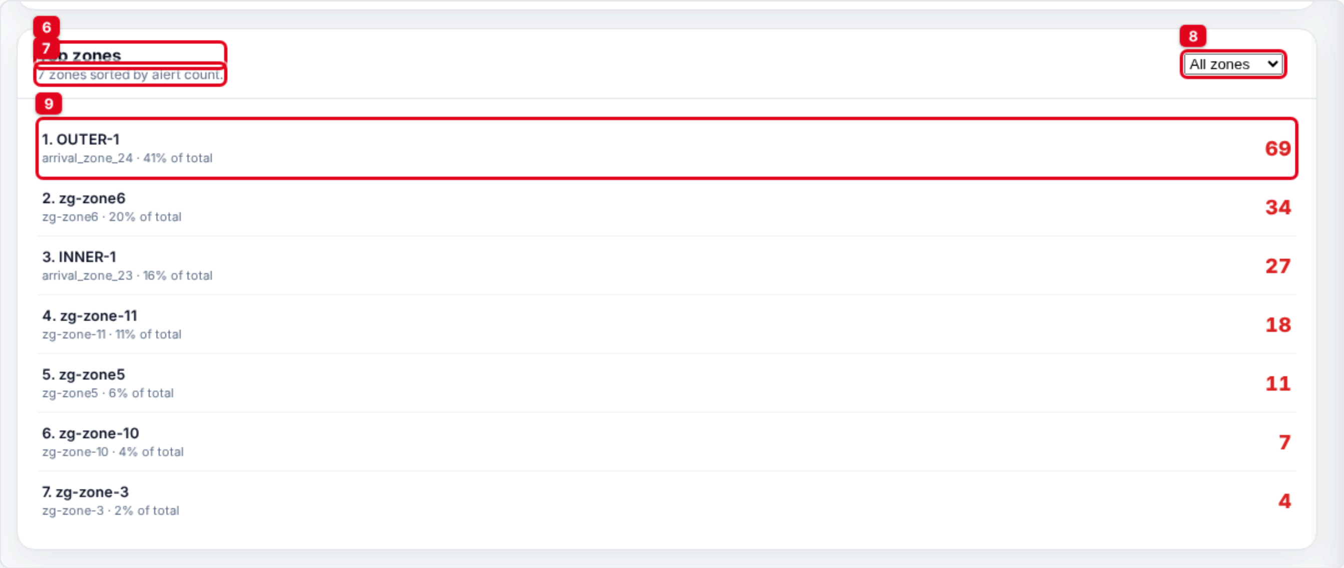
`#analytics` — recorded alert activity over time. Use it to spot patterns (busy hours, repeat-offender zones).

4.1 Controls and the trend chart



1. **Range** — the time window and bar granularity: **Last hour**, **Last 24h** (hourly bars), **Past 7 days**, **Past 30 days** (daily bars).
2. **Type** — which alert family to chart: **Zone capacity exceeding** (overcrowding) or **Vehicle overstay**.
3. **Chart title** — names the active series (e.g. *Zones exceeding capacity*).
4. **Total** — total alert count in the selected window and series (e.g. `170 alerts`).
5. **Bar chart** — one bar per hour or day. Bar height = alert count for that bucket (value printed above each bar). Y-axis = *Alerts*, X-axis = *Hour* (or *Day*).

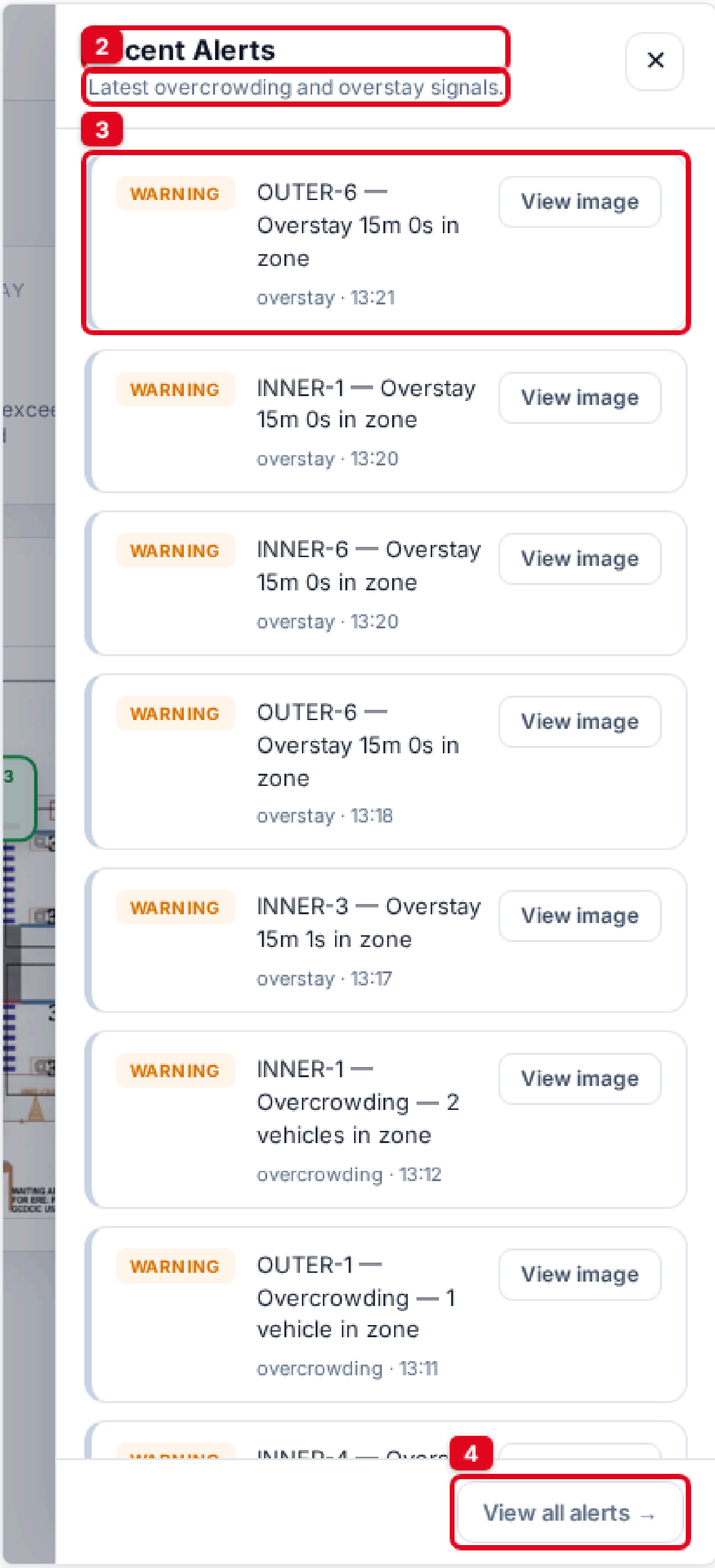
4.2 Top zones



1. **Panel title** — "Top zones".
2. **Subtitle** — how many zones are ranked, sorted by alert count.
3. **Zone filter dropdown** — restrict the whole Analytics view to one zone (or *All zones*).
4. **Ranked zone row** — for each zone: its **rank and name** (`1. OUTER-1`), the underlying **zone ID and share of total** (`arrival_zone_24 · 41% of total`), and the **alert count** for the window (red number on the right). A pager appears beneath the list when there are more than ten zones.

Notification bell & drawer

Available from the header on every page. Click the 🔔 **bell** to slide in the *Recent Alerts* drawer.



1. **Drawer title** — "Recent Alerts".
2. **Subtitle** — "Latest overcrowding and overstay signals."

3. **Recent alert card** — the most recent alerts (up to 15), each with its severity badge, headline, **View image** button, and type · time. This is a read-only quick-look; it mirrors the Alerts page.
4. **View all alerts** → — closes the drawer and opens the full **Alerts** page.

Close the drawer with the **✕** button or by clicking the dimmed background.

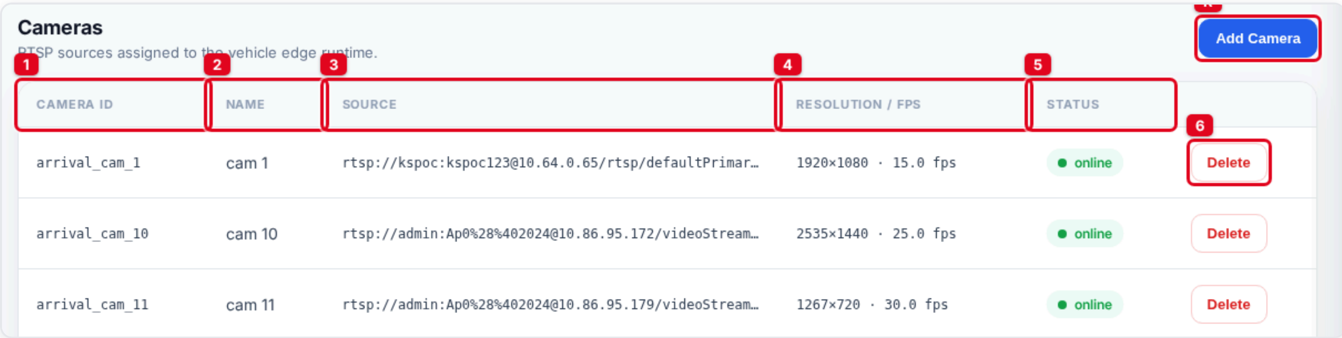
SECTION

Part B — Admin (Configuration)

These pages are visible only to the **admin** console (localhost). They change system configuration and expose raw diagnostics. Edits here directly affect what the edge runtime detects and what alerts fire.

5. Cameras

`#cameras` — the RTSP camera sources assigned to the vehicle edge runtime.

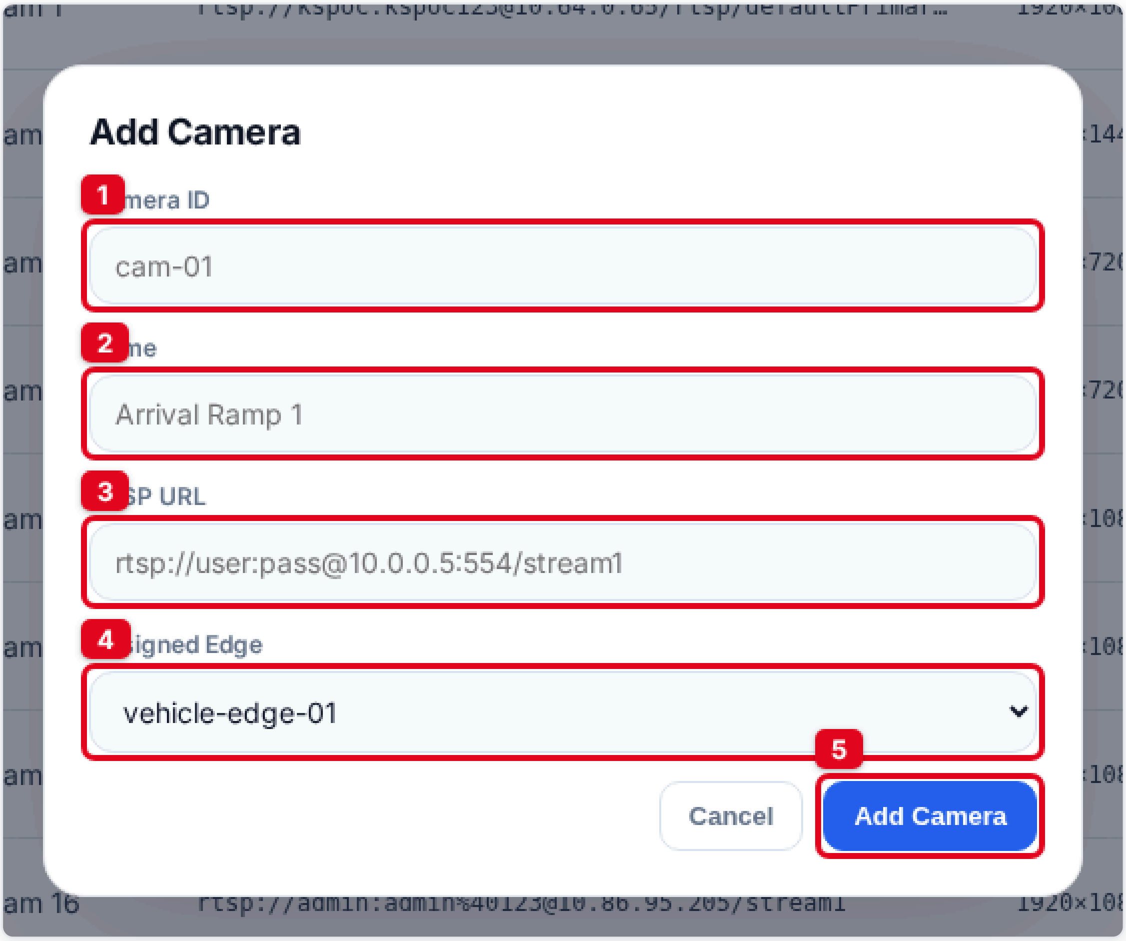


R. Add Camera — opens the *Add Camera* form (below).

Table columns:

1. **Camera ID** — the unique technical identifier (e.g. `arrival_cam_1`). Used everywhere else to link zones and alerts to this camera.
2. **Name** — the friendly label shown to operators (e.g. `cam 1`).
3. **Source** — the **RTSP URL** the edge pulls frames from (credentials included; truncated for display — hover for the full string).
4. **Resolution / FPS** — the negotiated stream resolution and frame rate, e.g. `1920×1080 · 15.0 fps`. Shows `–` until the edge reports it.
5. **Status** — `online` (green) when the edge is decoding the stream, otherwise `offline` / `error`.
6. **Delete** — removes the camera from the system. (Zones bound to it lose their feed — delete with care.)

5.1 Add Camera form



1. **Camera ID** *(required)* — unique ID, e.g. `cam-01`. Cannot clash with an existing camera.
2. **Name** — friendly display name, e.g. `Arrival Ramp 1`.
3. **RTSP URL** *(required)* — the full stream URL, e.g. `rtsp://user:pass@10.0.0.5:554/stream1`.
4. **Assigned Edge** — which edge runtime will run this camera (e.g. `vehicle-edge-01`).
5. **Add Camera** — saves the camera and pushes it to the edge. (**Cancel** discards.)

6. Zones

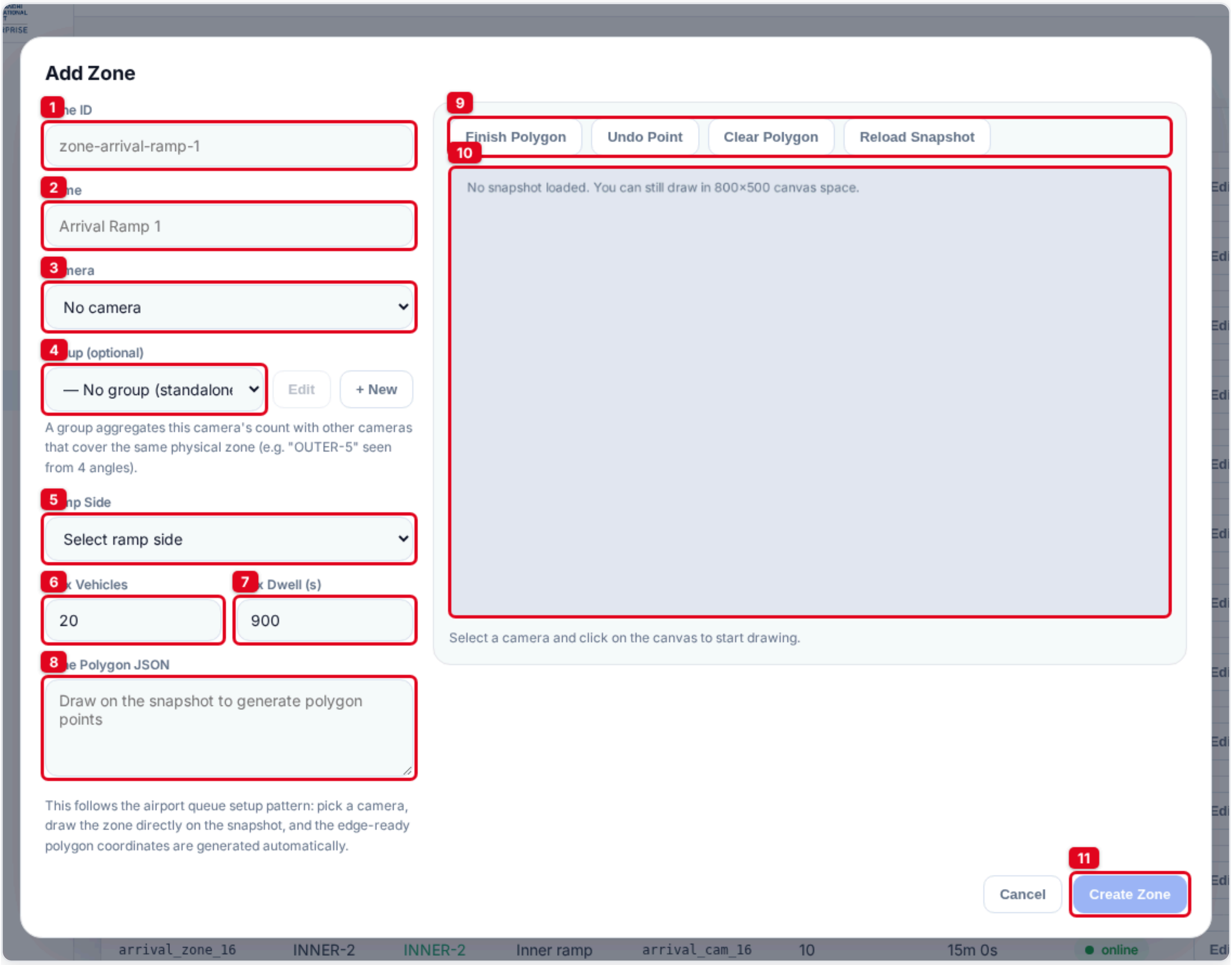
`#zones` — ramp geometry, occupancy thresholds, and dwell rules. A *zone* is a polygon drawn on one camera's view; vehicles inside it are counted and timed.

R. Add Zone — opens the zone editor (below).

Table columns:

1. **Zone ID** — unique technical identifier (e.g. `arrival_zone_1`).
2. **Name** — friendly label (e.g. `INNER-2`).
3. **Map slot** — which slot on the Overview Ramp Map this zone feeds:
4. a green `OUTER-n` / `INNER-n` = renders on the map at that slot;
5. **`– not mapped`** (grey italic) = the name doesn't match any floor-plan slot, so it won't appear on the map;
6. **`⚠ <slot>`** (red) = two zones claim the same slot — only one will render.
7. **Ramp Side** — **Inner ramp** or **Outer ramp**.
8. **Camera** — the camera this zone is drawn on (e.g. `arrival_cam_1`).
9. **Max Vehicles** — the occupancy threshold. Exceeding it = over capacity → overcrowding alert.
10. **Max Dwell** — the dwell-time threshold (e.g. `15m 0s`). A vehicle staying longer = overstay alert.
11. **Status** — `online` (zone active) or `offline` (inactive).
12. **Edit / Delete** — open the editor for this zone, or remove it.

6.1 Add / Edit Zone editor

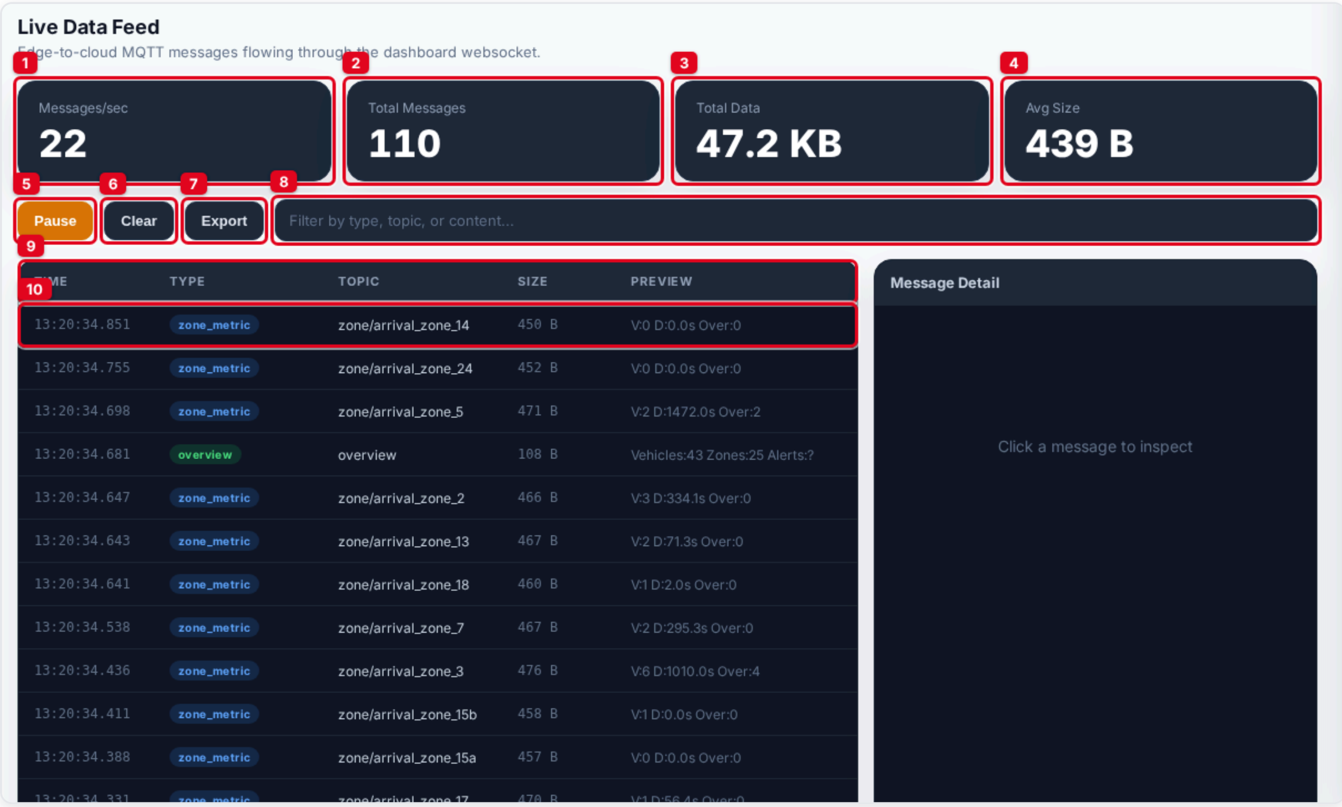


1. **Zone ID** *(required)* — unique ID, e.g. `zone-arrival-ramp-1`.
2. **Name** — friendly label. To appear on the Ramp Map, name it exactly `OUTER-1` … `OUTER-6` or `INNER-1` … `INNER-6`.
3. **Camera** — the camera whose view you'll draw on. Selecting it loads a live snapshot into the drawing canvas (10).
4. **Group (optional)** — attach this zone to a **zone group** so several cameras covering the *same* physical area are aggregated and alert once. Use **+ New** to create a group (Name, Group ID, and an optional **group Max Vehicles** that fires a single overcrowding alert for the summed count) or **Edit** to change the selected one.
5. **Ramp Side** — **Inner ramp** / **Outer ramp**.
6. **Max Vehicles** — occupancy threshold (default 20).
7. **Max Dwell (s)** — dwell threshold in **seconds** (default 900 = 15 min).
8. **Zone Polygon JSON** — the polygon coordinates. Read-only; it is generated automatically as you draw on the canvas.
9. **Polygon tools** — **Finish Polygon** (close the shape), **Undo Point** (remove the last vertex), **Clear Polygon** (start over), **Reload Snapshot** (refresh the camera still).

10. **Drawing canvas** — click on the camera snapshot to drop polygon points and outline the zone. (Drawing needs a mouse and a wide screen — it is disabled on mobile.)
11. **Create Zone / Save** — writes the zone and pushes the new geometry to the edge. (**Cancel** discards.)

7. Live Data Feed

`#feed` — the raw edge-to-cloud MQTT messages flowing through the dashboard WebSocket. A diagnostic / verification tool (admin-only).



Statistics (top):

1. **Messages/sec** — current throughput (averaged over a 5-second window).
2. **Total Messages** — cumulative messages received this session.
3. **Total Data** — cumulative bytes received.
4. **Avg Size** — average message size (Total Data ÷ Total Messages).

Toolbar:

1. **Pause / Resume** — freezes the live list so you can inspect a message (statistics keep counting). The button turns amber when paused.
2. **Clear** — empties the list and resets the session counters.
3. **Export** — downloads the current buffered messages as a timestamped JSON file.
4. **Filter** — free-text filter over message type, topic, and content.

Message table:

1. **Column headers** — **Time** (HH:MM:SS.mmm), **Type**, **Topic**, **Size**, **Preview**.

2. **Message row** — one MQTT message. The **Type** badge is colour-coded: `zone_metric` (blue, per-zone counts), `overview` (green, ramp-wide rollup), `alert` (red), `subscribed/connected` (system), and muted greys for disconnect/error events. The **Preview** summarises the payload — for a `zone_metric`: `V:<vehicles> D:<avg dwell> Over:<overstays>`. Click any row to inspect it.

7.1 Message Detail

Click a row (pause first to stop the list scrolling) to see the full message on the right.

1. **Time** — exact arrival time of the message.
2. **Size** — message size in bytes.
3. **Type** — the message type badge (e.g. `zone_metric`).
4. **Topic** — the MQTT topic, e.g. `zone/arrival_zone_3b`.
5. **Full Payload** — the complete decoded JSON. For a `zone_metric` this includes the edge ID, camera ID, timestamp, `vehicle_count`, `vehicle_count_by_type`, `max_vehicles`, `occupancy_pct`, `overstay_count`, `avg_dwell_time_s`, `max_dwell_time_s`, `total_entered`, and more — the raw numbers that drive every other screen in the dashboard.

Appendix — quick reference

Alert types

Type	Raised when	Severity
Overcrowding	A zone's vehicle count exceeds its **Max Vehicles**	warning / critical
Overstay	A vehicle stays longer than the zone's **Max Dwell**	warning / critical
Camera offline	The edge stops receiving frames from a camera	warning / critical
Camera recovered	A previously offline camera resumes	info

Occupancy colour bands (zones / map / KPIs)

Band	Condition	Colour
Calm	occupancy ≤ 50 %	green
Busy / Near capacity	> 50 % and within the limit	amber
Over capacity	count strictly over Max Vehicles	red

Edge resource colours

Utilisation	Colour
< 55 %	green
55 – 85 %	amber
≥ 85 %	red

Access summary

Page	Operator (LAN viewer)	Admin (server console)
Overview, Live View, Alerts, Analytics	✅	✅
Data Feed	❌	✅
Cameras, Zones, Edge Status	❌	✅